

POLICY AND REINFORCEMENT LEARNING BASED MISSION RESILIENCE SIMULATION FOR LUNAR SOUTH POLE ROVER NAVIGATION

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2. Abstract

This report presents a policy and reinforcement learning (RL) based mission resilience simulation for a solar-powered lunar rover operating near the south pole of the Moon. The study addresses the question of how robust different rover navigation policies are under terrain risk, shadow risk, sensor uncertainty, blocked paths, limited energy, and communication delay conditions. A 200 x 200 grid-world was constructed at 25 m per cell resolution, representing a 5 km x 5 km study area, using four raster datasets: LOLA Digital Elevation Model (DEM) for terrain slope, LROC WAC shadow mask for illumination risk, DFSAR Circular Polarisation Ratio (CPR) for data confidence, and MiniRF S1 for science reward.

Four baseline navigation policies were implemented using the A* search algorithm: Shortest Path, Safest Path, Science Priority, and Uncertainty Aware. A Q-Learning policy was trained offline over 150,000 episodes and evaluated separately. All five policies were tested under six mission failure scenarios. Performance was measured using success rate, energy consumed, science return, hazard exposure, and policy compute cost. The Safest Path and Uncertainty Aware policies achieved the highest resilience score (83.3% success across scenarios). Q-Learning demonstrated the lowest per-step execution cost ($O(1)$, 39 KB policy map) but the lowest overall resilience (50%) due to its inability to replan when paths are blocked. These results demonstrate a clear trade-off between onboard execution simplicity and mission adaptability.

3. Introduction

The lunar south pole is among the most scientifically and strategically important targets in current space exploration. Permanently shadowed regions (PSRs) in its crater floors are hypothesised to contain water ice, volatiles, and other resources relevant to both science and long-duration human presence. Missions such as ISRO Chandrayaan-3 and the upcoming NASA VIPER rover are designed specifically to characterise these regions. However, the south polar environment presents severe navigation challenges: steep crater walls, permanently shadowed terrain, extreme temperature gradients, and limited solar illumination for power generation.

A digital twin framework for lunar south-polar operations requires at least four task-specific input layers: terrain risk derived from topographic slope, shadow risk derived from illumination conditions, data confidence derived from radar coverage, and science reward derived from radar response proxies for surface composition. These layers must be integrated into a grid-world simulation environment that can test different rover decision policies and quantify mission resilience.

This project builds such a simulation and evaluates five navigation policies under realistic failure conditions. The primary motivation is to identify which policy offers the best balance between onboard computational feasibility and mission robustness. The work is designed to be later integrated with a broader digital twin pipeline developed by a parallel intern team working on sensor fusion, confidence classification, and science target accessibility.

4. Research Question and Objectives

Research Question: How robust is a rover mission under different terrain, uncertainty, and failure conditions?

Objectives:

1. To construct a grid-world simulation environment from real lunar south-polar raster datasets at 25 m per cell resolution.
2. To implement four A*-based baseline navigation policies representing different optimisation criteria (distance, terrain safety, science return, and data confidence).
3. To train a Q-Learning policy offline and evaluate whether its compact policy map is feasible for onboard rover execution.
4. To test all five policies under six mission failure scenarios and measure resilience using quantitative metrics.
5. To identify limitations in the simulation including terrain normalisation inconsistencies, disconnected grid components, and differences in science metric measurement between policy types.

This project does not attempt to perform multi-sensor data fusion, generate global confidence classification maps, validate science target accessibility, or simulate real-time onboard processing. These are addressed by parallel intern projects.

5. Dataset Description

Four raster datasets were used to construct the policy input layers for the grid-world simulation. Table 1 summarises each dataset.

Table 1: Dataset Summary

Dataset	Product Used	Resolution	What It Represents	How Used in This Project
LOLA	DEM (elevation raster)	25 m/cell (native ~30 m, resampled)	Lunar surface elevation	Slope calculated from elevation; slope used to compute terrain risk and passability threshold
LROC WAC	Shadow mask (5 m, resampled to 25 m)	25 m/cell after resampling	Solar illumination fraction per cell (0 = lit, 1 = dark)	Shadow risk layer for movement cost and scenario testing
DFSAR	Circular Polarisation Ratio (CPR)	25 m/cell	Radar backscatter polarisation ratio; proxy for surface roughness	Primary component (weight 0.6) of data confidence layer and science reward layer
MiniRF	S1 (first Stokes parameter)	25 m/cell	Total radar backscatter power; proxy for surface composition variation	Secondary component (weight 0.4) of data confidence layer and science reward layer

Limitations per dataset:

- LOLA: Slope was normalised using slope / 25 in the movement cost formula. An earlier documentation version stated slope / 15. This inconsistency was discovered during project review and is reported as a known limitation. The passability threshold was corrected from 15° to 20° after an initial disconnected terrain diagnosis (Section 8).
- LROC: The shadow mask was resampled from 5 m to 25 m using nearest-neighbour interpolation. Sub-cell shadow variation is not captured. The study area contains 30.7% predominantly shadowed cells.
- DFSAR and MiniRF: Both datasets were used as proxies. DFSAR CPR does not directly measure ice presence. MiniRF S1 does not directly measure compositional type. Their use as science reward proxies is provisional.
- An earlier study_area_crop_info.txt file in the project outputs folder was from an 80 x 80 grid run at 60 m resolution and was never regenerated for the current 200 x 200 grid. This file should be regenerated for final submission.

6. Methodology

Step 1: Reference Rover Definition

Purpose: Establish the rover hardware and constraint assumptions that govern all policy rules.

Input: Published specifications for ISRO Chandrayaan-3 Pragyan and NASA VIPER rover concepts.

Method: Key parameters were extracted from public mission documentation. Where exact values were unavailable, conservative assumptions were declared and recorded as simulation parameters rather than verified facts.

Output: Reference rover architecture document. Key parameters used in this simulation: solar power dependency (no operation in permanent shadow), energy budget of 650 cost units per traverse, 8-directional movement, and an onboard computer assumed to support at most $O(1)$ per-step policy lookup without in-flight replanning.

Step 2: Rover Hardware and Constraint Recording

Purpose: Translate rover assumptions into quantitative simulation constraints.

Method: The energy budget was set to 650 cost units representing a full single-traverse energy allocation. The passability constraint was defined as slope ≤ 20 degrees. An impassability threshold (IMPASS_THR) of 0.90 was applied: any cell with terrain risk ≥ 0.90 was marked impassable. Science reward cells were defined at the 95th percentile of passable cell science reward values.

Step 3: Raster File Reading and Layer Extraction

Purpose: Read and understand the four input raster files and extract values for the grid.

Input: LOLA DEM, LROC shadow mask, DFSAR CPR, MiniRF S1 rasters.

Method: Python with rasterio was used to read each raster. Metadata including resolution, no-data value, and value range were recorded. Because the source raster files were at non-uniform resolutions, all layers were resampled or extracted to match the 200 x 200 grid at 25 m per cell. For intermediate steps of the project where source rasters were unavailable, numpy arrays were extracted from rendered PNG risk map images via colourmap inversion. This is recorded as a known limitation.

Step 4: Study Area Selection

Purpose: Choose a 5 km x 5 km region near the lunar south pole for the simulation.

Input: Full-resolution raster tiles covering the lunar south polar region.

Method: A 200 x 200 cell crop was extracted from each raster, representing a 5 km x 5 km area at 25 m per cell. The start cell was set at grid position (0, 199) — upper left area — and the goal cell at (199, 0) — lower right area. The area was selected to include a mix of passable and impassable terrain and was verified to contain a connected path between start and goal after the slope threshold was corrected to 20 degrees.

Step 5: Preparation of Policy Input Layers

Purpose: Generate the four task-specific risk and reward layers used by all policies.

Method and Output:

- **Terrain Risk:** Slope in degrees was calculated from the LOLA DEM using a standard 3x3 gradient kernel. Terrain risk was normalised as $tr = \text{slope_deg} / 25$, capped at 1.0. Cells with $tr \geq 0.90$ were marked impassable. This produced `terrain_risk_map.png`.
- **Shadow Risk:** The LROC shadow fraction per cell was used directly as shadow risk (sh), ranging from 0 = fully lit to 1 = fully dark. This produced `shadow_risk_map.png`.
- **Data Confidence:** A weighted combination of DFSAR coverage (weight 0.6) and MiniRF S1 normalised values (weight 0.4) was used: $cn = 0.6 * \text{dfsar} + 0.4 * \text{minirf}$. Mean confidence across the grid is 0.704. This produced `confidence_map.png`.
- **Science Reward:** A weighted combination of DFSAR CPR (weight 0.6) and MiniRF S1 (weight 0.4) was used: $sr = 0.6 * \text{dfsar_cpr} + 0.4 * \text{minirf_s1}$. High-reward cells ($sr > 0.7$) total 902 across the full grid; 12 of these are accessible (traversable). Science cells for the bonus threshold were defined at the 95th percentile of passable cells. This produced `science_reward_map.png`.

Step 6: Grid-World Construction

Purpose: Convert the study area into a simulation environment.

Method: Each (row, col) index of the 200 x 200 array became a possible rover state. Each cell stored its terrain risk, shadow risk, confidence, science reward, passability flag, and precomputed movement cost to all eight neighbours. The grid was exported as `grid_world.csv` and `grid_cell_properties.csv`.

Passability: 34,624 cells (86.6%) are passable. 5,376 cells (13.4%) are impassable.

Step 7: Action Space Definition

Purpose: Define how the rover can move within the grid.

Method: Eight actions were defined — Up, Down, Left, Right, Up-Right, Up-Left, Down-Right, Down-Left — corresponding to movement in all eight grid directions. Cardinal moves have unit cost multiplier 1.0; diagonal moves use a cost multiplier of $\sqrt{2}$ to represent the longer physical distance. Movement outside the grid boundary and into impassable cells was prohibited. The action space was saved to `action_space.txt`.

Step 8: Reward Function Definition

Purpose: Define how the rover receives reward and penalty for each action.

Method: The movement cost formula, goal reward, science bonus, and energy budget constraint are described in full in Section 7. The reward function was saved to `reward_function.txt` and operating constraints were saved to `operating_constraints_table.csv`.

Step 9: Baseline Policy Implementation

Purpose: Test four rule-based A* navigation policies before RL training.

Method: Each policy was implemented as a modified A* search using a different edge cost function:

- **Shortest Path:** edge cost = diagonal multiplier only (minimize steps).
- **Safest Path:** edge cost = movement cost formula with terrain and shadow weighted equally (minimize hazard exposure).
- **Science Priority:** movement cost formula with a science discount subtracted from edge cost to reward visiting high-science cells.

- **Uncertainty Aware:** movement cost formula with an additional (1 - confidence) penalty term to avoid low-confidence cells.

An earlier version of the Uncertainty Aware policy contained a bug where its cost function was identical to Shortest Path due to an incorrect weight assignment. This was diagnosed, corrected, and the policy was rerun. Results in this report reflect the corrected implementation.

All four policies produced valid paths from start (0,199) to goal (199,0) within the energy budget of 650 units. Results were saved to `baseline_policy_results.csv` and `baseline_policy_maps.png`.

Step 10: Q-Learning Training

Purpose: Train a simple RL policy offline for potential onboard deployment.

Method: A tabular Q-learning algorithm was implemented. Each state was a (row, col) grid cell index. The Q-table had $200 \times 200 \times 8 = 320,000$ entries. Training used epsilon-greedy exploration decaying from 1.0 to 0.01. The reward signal used the movement cost formula as step penalty, plus a large positive goal reward. Science reward was excluded from training to prevent the rover from loitering at high-science cells instead of progressing to the goal. Science was only measured in evaluation. Training ran for 150,000 episodes with discount factor $\gamma = 0.99$ and learning rate $\alpha = 0.1$. Training completed in 300 seconds. The learned greedy action map (argmax over Q-table per cell) was saved as a compact uint8 array of 39 KB.

Step 11: Policy Compute Cost Evaluation

Purpose: Assess the feasibility of each policy for constrained onboard rover execution.

Method: Planning time (ms), nodes explored, algorithmic complexity class, storage size (KB), and onboard suitability were recorded for each policy. A* policies require ground-side planning and uplink of waypoint lists (approximately 900 to 1,100 bytes per mission). Q-Learning requires only a 39 KB uint8 policy map lookup per step.

Step 12: Mission Failure Scenario Tests

Purpose: Evaluate policy resilience under six realistic failure modes.

Method: Each scenario was applied to the grid environment and all five policies were tested:

- **S1 Sensor Dropout:** 25% of non-impassable cells had their confidence value dropped to 0, simulating sensor failure. Policies rerouted using remaining valid cells.
- **S2 Blocked Path:** A horizontal band of cells was added to the impassable set, forcing all policies to find alternative routes. A* policies replanned successfully. Q-Learning failed because its frozen policy map had no mechanism to avoid newly blocked cells.
- **S3 Limited Energy:** The energy budget was reduced from 650 to 430 cost units. Policies whose nominal energy use exceeded 430 failed.
- **S4 Wrong Resource Estimate:** Science reward values were perturbed by additive Gaussian noise ($\sigma = 0.1$). All policies reached the goal but Science Priority deviated from its optimal route toward noise-inflated cells.
- **S5 Increased Terrain Uncertainty:** Terrain risk values were perturbed by multiplicative noise (factor = 1.1). All policies remained robust.
- **S6 Communication Delay:** A five-step communication delay penalty was applied, adding overhead steps and proportional energy cost. All policies that could complete under normal conditions still completed, but with increased step count and energy.

Step 13 and 14: Resilience Measurement and Policy Comparison

Purpose: Calculate final metrics and compare all five policies.

Method: Success was binary: a policy passed a scenario if it reached the goal within the applicable energy budget. Success rate was computed as the fraction of the six scenarios passed. All other metrics (steps, energy, science, hazard exposure) were recorded per scenario and aggregated.

7. Equations, Rules, and Thresholds

7.1 Terrain Risk

Equation name: Terrain risk normalisation

$$tr(c) = \min(\text{slope_deg}(c) / 25, 1.0)$$

Variables: $\text{slope_deg}(c)$ = slope in degrees at cell c , computed from LOLA DEM using 3x3 gradient kernel. Result is clamped to $[0, 1]$.

Reason: Provides a continuous risk score for movement cost calculation. Cells with slope $> 25^\circ$ receive $tr = 1.0$.

Known inconsistency: Earlier project documentation stated slope / 15 as the normalisation denominator. The implemented code uses slope / 25. This discrepancy is reported as a limitation and should be standardised before integration with the digital twin.

7.2 Shadow Risk

Equation name: Shadow risk

$$sh(c) = \text{LROC_shadow_fraction}(c) \quad [0 = \text{fully lit}, 1 = \text{fully dark}]$$

Reason: Direct use of the shadow mask fraction as a per-cell risk value. No additional normalisation is required.

7.3 Data Confidence

Equation name: Confidence fusion

$$cn(c) = 0.6 * \text{dfsar_coverage}(c) + 0.4 * \text{minirf_s1_norm}(c)$$

Variables: dfsar_coverage is binary (0 or 1) radar coverage. minirf_s1_norm is the MiniRF S1 value normalised to $[0, 1]$. Grid mean confidence = 0.704.

Reason: DFSAR coverage is treated as the more reliable source (weight 0.6). MiniRF provides supplementary coverage context (weight 0.4).

7.4 Science Reward

Equation name: Science reward fusion

$$sr(c) = 0.6 * \text{dfsar_cpr}(c) + 0.4 * \text{minirf_s1}(c)$$

Variables: dfsar_cpr = DFSAR circular polarisation ratio normalised to $[0, 1]$. minirf_s1 = MiniRF first Stokes parameter normalised to $[0, 1]$. Science bonus threshold = 95th percentile of sr over all passable cells.

Reason: Both datasets are radar-based proxies. CPR is given higher weight as it is more directly related to surface roughness anomalies associated with ice deposits.

7.5 Movement Cost

Equation name: Movement cost per step

$$\text{cost}(c, d) = d * (1 + 0.6 * tr(c) + 0.3 * sh(c) + 0.1 * (1 - cn(c)))$$

Variables: d = direction multiplier (1.0 for cardinal moves, $\sqrt{2} = 1.414$ for diagonal moves). $tr(c)$, $sh(c)$, $cn(c)$ as defined above. Weights sum to 1.0: terrain 0.6, shadow 0.3, confidence 0.1.

Assumption: Terrain safety is the dominant cost factor for a solar-powered rover operating on steep south-polar terrain. Shadow risk is secondary because it affects both power generation and thermal management. Confidence uncertainty is tertiary.

7.6 Passability Rule

```
passable(c) = True  if  tr(c) < IMPASS_THR  and  slope_deg(c) <= 20
               False otherwise
IMPASS_THR = 0.90
```

Reason: A cell with terrain risk ≥ 0.90 is treated as impassable. The initial threshold was slope ≤ 15 degrees, which produced a disconnected grid (the goal cell was unreachable). After diagnosis, the threshold was raised to 20 degrees, yielding 34,624 passable cells and a fully connected traversal region.

7.7 Science Priority Cost Discount

Equation name: Science A* edge cost

```
cost_sci(c, d) = cost(c, d) - lambda_sci * max(0, sr(c) - science_threshold)
```

Variables: λ_{sci} = science discount weight. $science_threshold$ = 95th percentile of passable cell science rewards. The discount is only applied when $sr(c)$ exceeds the threshold, preventing over-discounting of all cells.

7.8 Uncertainty Aware Cost Penalty

Equation name: Uncertainty Aware A* edge cost

```
cost_ua(c, d) = cost(c, d) + mu_unc * (1 - cn(c))
```

Variables: μ_{unc} = uncertainty penalty weight. $(1 - cn(c))$ is the uncertainty at cell c .

7.9 Q-Learning Update Rule

Equation name: Q-value update (Bellman equation)

```
Q(s, a) <- Q(s, a) + alpha * [r + gamma * max_{a'} Q(s', a') - Q(s, a)]
```

Variables: $\alpha = 0.1$ (learning rate), $\gamma = 0.99$ (discount factor), r = immediate reward, s' = next state after action a from state s .

7.10 Reward Function

The step reward used in Q-Learning training is:

```
r(s, a) = R_GOAL          if s' = goal cell
         - cost(c, d)      if move is valid
         - R_FAIL         if energy budget exceeded or loop detected
```

R_GOAL was set to a large positive value calibrated to outweigh the maximum accumulated step penalty over a full 300-step path. Science reward was excluded from training ($science_in_training = False$) to prevent farming behaviour. Science was measured during evaluation only, as the cumulative sum of $sr(c)$ over all first-visit cells where $sr > 0.5$.

7.11 Energy Budget Rule

energy_used = sum of cost(c_i, d_i) for all steps i in the path

Mission success requires: energy_used <= energy_budget (650 cost units, nominal)

8. Validation and Checks

8.1 Grid Connectivity Check

What was checked: Whether a valid path exists from start (0,199) to goal (199,0) after applying the passability threshold.

How it was checked: A breadth-first search was run from the start cell. The goal cell was checked for membership in the reachable component.

Result: Under the initial slope ≤ 15 degree threshold, the goal cell was unreachable. The terrain in the upper-right portion of the grid formed an impassable barrier at that threshold. The threshold was corrected to slope ≤ 20 degrees, which produced 34,624 passable cells and a fully connected traversal corridor.

What it means: The corrected threshold is physically reasonable for a rover with VIPER-class wheel design. The change is documented as a design decision.

8.2 Energy Budget Feasibility Check

What was checked: Whether all four baseline policies complete within the 650-unit energy budget under nominal conditions.

Result: All four A* policies and Q-Learning complete within budget (max energy 435.8 for Shortest Path). All passed the nominal run.

8.3 Science Threshold Sensitivity Check

What was checked: Whether the science bonus threshold was too restrictive to affect policy paths.

Result: An initial 99th percentile threshold produced zero science cells in accessible areas, making the Science Priority policy identical to Shortest Path. The threshold was corrected to the 95th percentile, enabling a meaningful science bonus.

8.4 Q-Learning Convergence Check

What was checked: Whether Q-Learning reached a stable solution.

Result: 100% training win rate and 100% evaluation win rate were achieved over 150,000 episodes. The policy did not show degradation in later episodes. Training time was 300 seconds.

8.5 Policy Compute Cost Comparison

What was checked: Whether each policy is computationally feasible for an onboard-constrained rover.

Result: Q-Learning step lookup is 0.76 microseconds, using $O(1)$ complexity and 39 KB storage. A* policies require 223 to 424 ms offline planning and $O(N \log N)$ complexity. All five policies were evaluated for onboard suitability. See Table 3 and Section 9.3.

8.6 Uncertainty Aware Policy Bug Check

What was checked: Whether the Uncertainty Aware policy cost function was correctly distinguishable from Shortest Path.

Result: The original implementation used an incorrect weight assignment that produced the same edge cost as Shortest Path. After correcting the confidence penalty term, the Uncertainty Aware

policy produced a distinct path with lower hazard exposure (103.1 terrain exposure vs 121.5 for Shortest Path) and higher science return (77.25 vs 57.19).

9. Results

9.1 Input Layer Maps

Figure 1 shows the four policy input layers: terrain risk, shadow risk, confidence, and science reward. These layers were derived from the four raster datasets described in Section 5. The terrain risk map (upper left panel of Figure 1) shows that the study area contains an impassable ridge running diagonally across the upper third of the grid. The shadow risk map shows 30.7% of cells are predominantly shadowed, concentrated in the upper portion of the grid. The confidence map shows that most of the grid has moderate to high confidence (mean 0.704), with a band of low-confidence cells in the upper rows. The science reward map shows high-reward cells ($sr > 0.7$) distributed primarily in the upper-left quadrant of the grid.

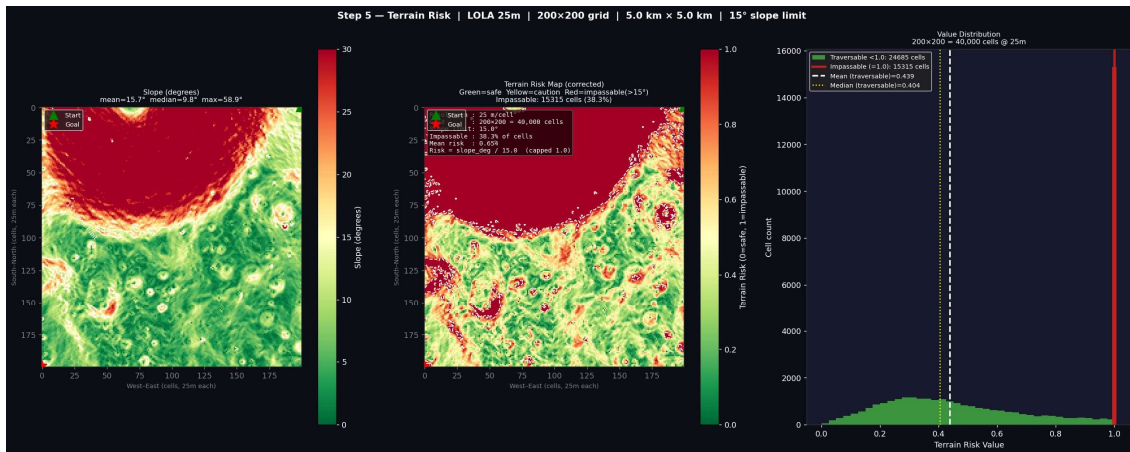


Figure 1a: Terrain Risk Map. Left: slope in degrees from LOLA DEM. Centre: terrain risk (0 = safe, 1 = impassable, red > 15 degree cells). Right: distribution of terrain risk values across passable and impassable cells. 38.3% of cells are impassable under the initial slope ≤ 15 degree threshold shown; this was subsequently corrected to 20 degrees.

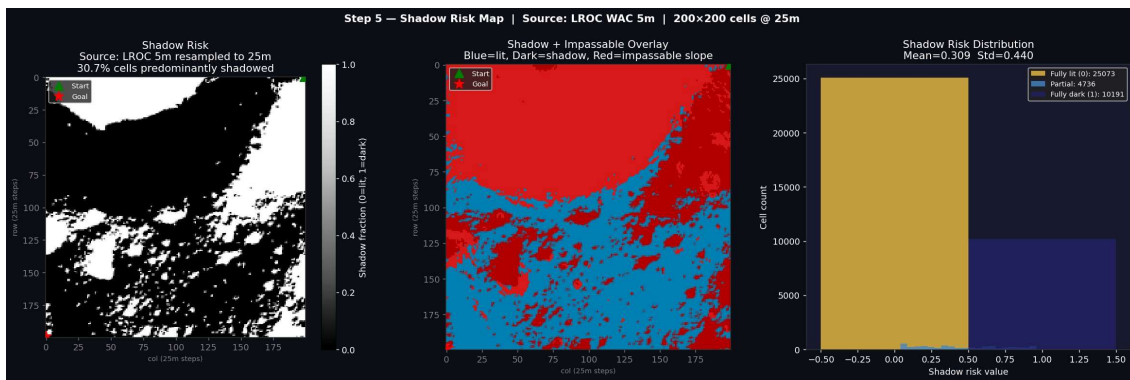


Figure 1b: Shadow Risk Map. Left: shadow fraction per cell from LROC WAC. Centre: overlay with impassable cells (red = impassable slope, blue = lit, dark = shadow). Right: distribution showing 25,073 fully lit cells, 4,736 partial, and 10,191 fully dark cells.

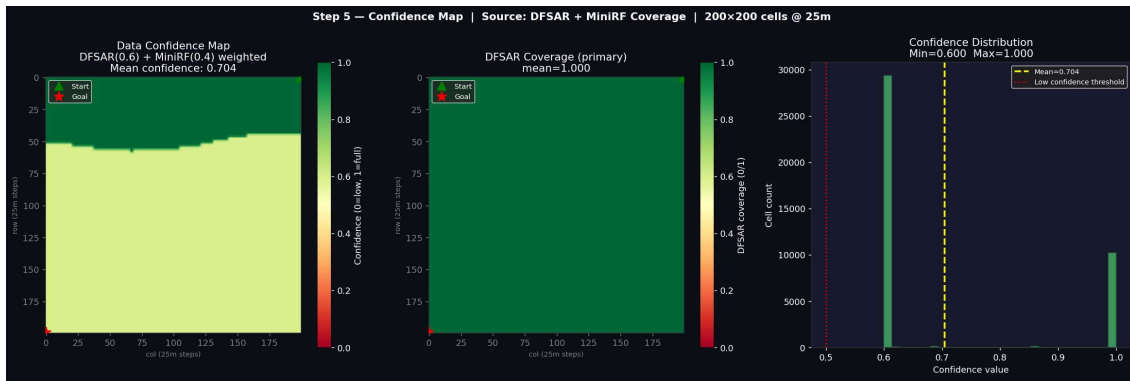


Figure 1c: Data Confidence Map. Left: combined DFSAR (0.6 weight) and MiniRF (0.4 weight) confidence, mean = 0.704. Centre: DFSAR coverage only (near-complete at mean = 1.0). Right: distribution showing the majority of cells at confidence 0.6 or 1.0.

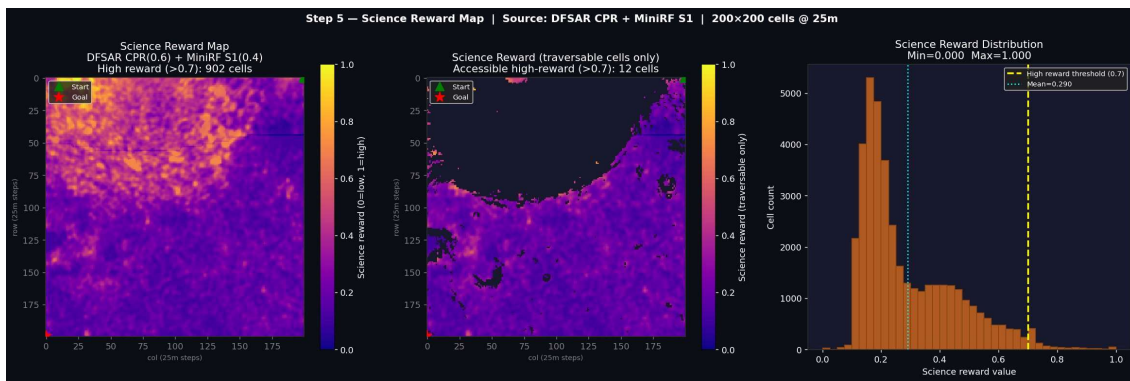


Figure 1d: Science Reward Map. Left: full-grid science reward (DFSAR CPR 0.6 + MiniRF S1 0.4). Centre: science reward for traversable cells only; 12 cells exceed the 0.7 high-reward threshold and are accessible. Right: distribution across traversable cells, mean = 0.290.

9.2 Baseline Policy Results

Table 2 presents the nominal performance of all four A* baseline policies and the Q-Learning policy. Figure 2 shows the four baseline policy paths overlaid on the terrain risk map.

Table 2: Baseline Policy Nominal Performance

Policy	Steps	Distance (m)	Energy Used	Terrain Exp.	Shadow Exp.	Science Return	Hazard Cells
Shortest Path	229	7,516	435.8 / 650	121.5	84.6	57.19	111
Safest Path	273	8,140	418.9 / 650	110.6	12.4	74.53	79
Science Priority	255	7,856	412.0 / 650	109.2	19.6	73.06	78
Uncertainty Aware	260	8,033	429.9 / 650	103.1	62.8	77.25	83
Q-Learning (eval)	268	—	440.5 / 650	—	—	26.5*	—

*Q-Learning science return is measured as cumulative sr over first-visit cells with sr > 0.5. Baseline science return is the cumulative sum of sr(c) over all path cells. These metrics are not directly comparable; see Section 10.

Key observations from Table 2:

- Safest Path uses the least energy (418.9) despite taking the longest path (273 steps). This occurs because it avoids high-terrain-risk cells, which are the dominant term in the movement cost formula. Shorter paths that traverse steep cells accumulate higher per-step cost.
- Uncertainty Aware collects the most science (77.25) despite not being designed as a science policy. High-confidence cells in this grid spatially overlap with high-science-reward terrain.
- Shortest Path incurs the highest terrain exposure (121.5) and shadow exposure (84.6) because it minimises step count without penalising hazardous terrain.

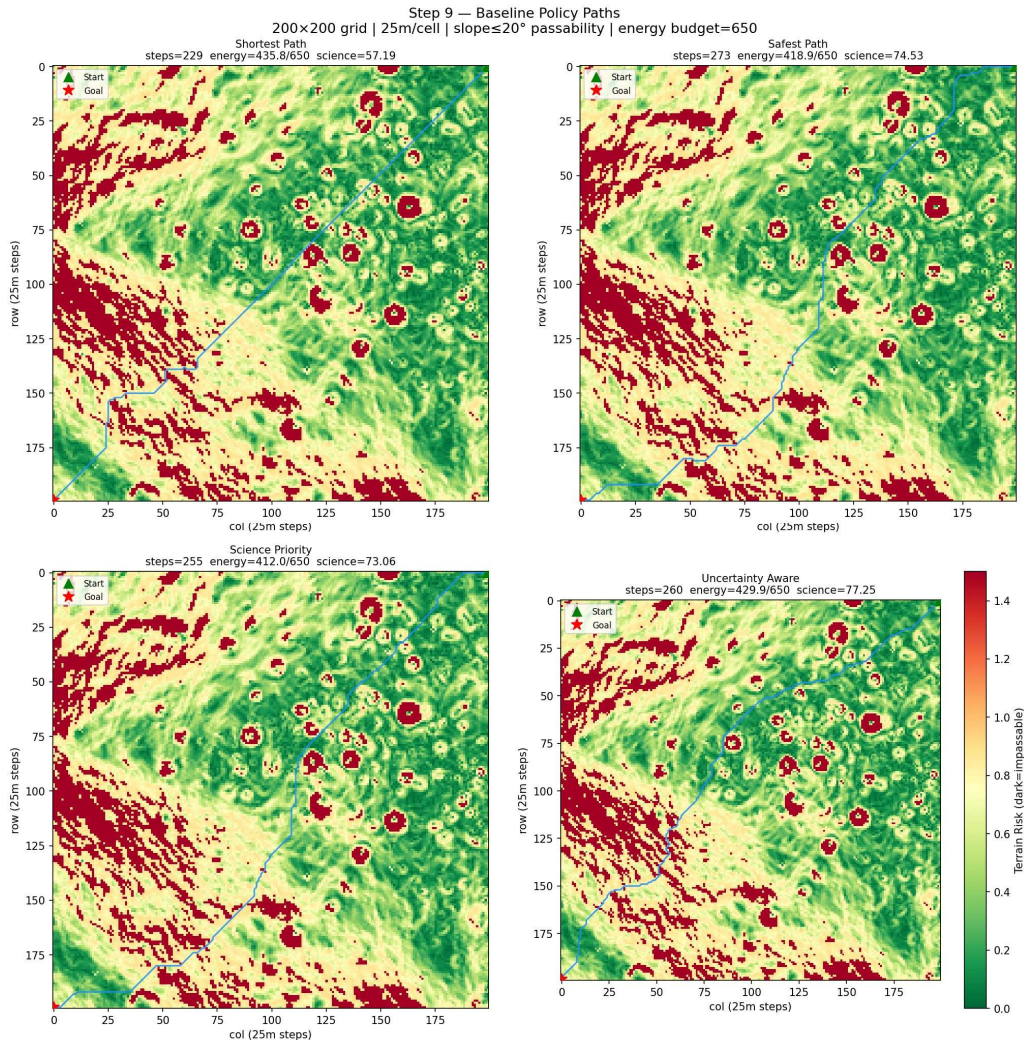


Figure 2: Baseline Policy Paths. Each panel shows the path (blue line) overlaid on the terrain risk map (green = low risk, red = impassable). Start is upper left (triangle), Goal is lower right (star). Path statistics are shown in each title.

9.3 Policy Compute Cost

Table 3 shows the compute and storage requirements for each policy.

Table 3: Policy Compute Cost Comparison

Policy	Plan Time (ms)	Nodes Explored	Complexity	Storage (KB)	Execution Mode	Onboard Suitable
Shortest Path	223	38,476	$O(N \log N)$	234.4 (waypoints)	Offline plan + onboard follow	Yes (uplink waypoints)
Safest Path	424	49,539	$O(N \log N)$	234.4 (waypoints)	Offline plan + onboard follow	Yes (uplink waypoints)
Science Priority	361	59,103	$O(N \log N)$	234.4 (waypoints)	Offline plan + onboard follow	Yes (uplink waypoints)
Uncertainty Aware	404	52,692	$O(N \log N)$	234.4 (waypoints)	Offline plan + onboard follow	Yes (uplink waypoints)
Q-Learning	0.00076 ms/step	1 (lookup)	$O(1)$	39 KB (policy map)	Fully onboard	Yes (ideal)

The Q-Learning policy map (Figure 3) shows the greedy action at every passable cell. The majority of the grid (blue region) directs the rover upward toward the goal. Coloured bands near the traversal corridor reflect diagonal and lateral movements used to navigate around impassable terrain clusters.

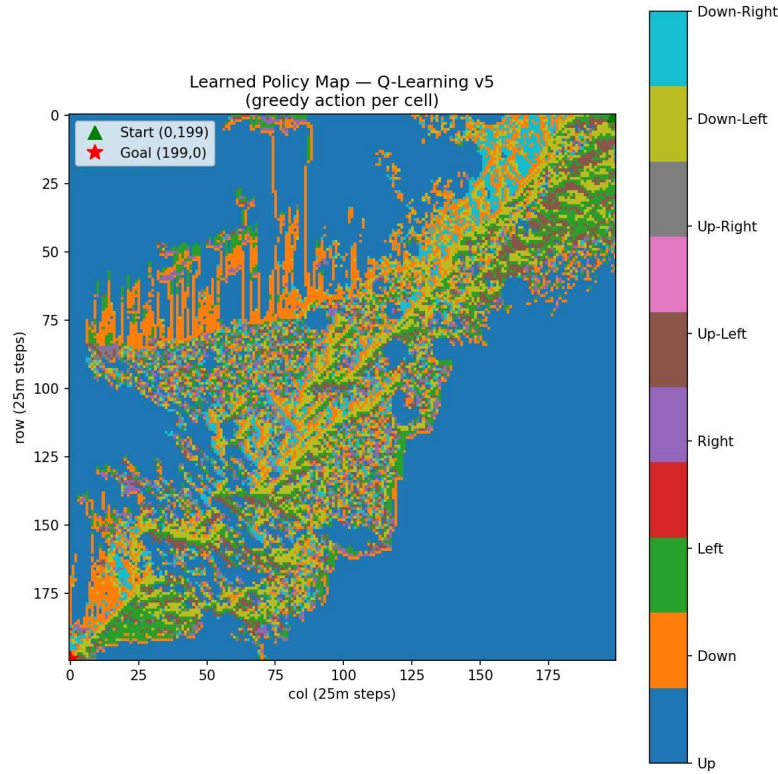


Figure 3: Q-Learning Learned Policy Map. Each coloured cell shows the greedy action ($\text{argmax } Q$) at that grid position. Blue = Up (dominant direction toward goal). Orange = Down. Other colours represent diagonal

and lateral actions in the traversal corridor. Impassable cells retain the initialised Q-value direction and are not visited during evaluation.

9.4 Scenario Test Results

Figure 4 shows the scenario test results for all six failure scenarios across all five policies. Table 4 presents the success summary.

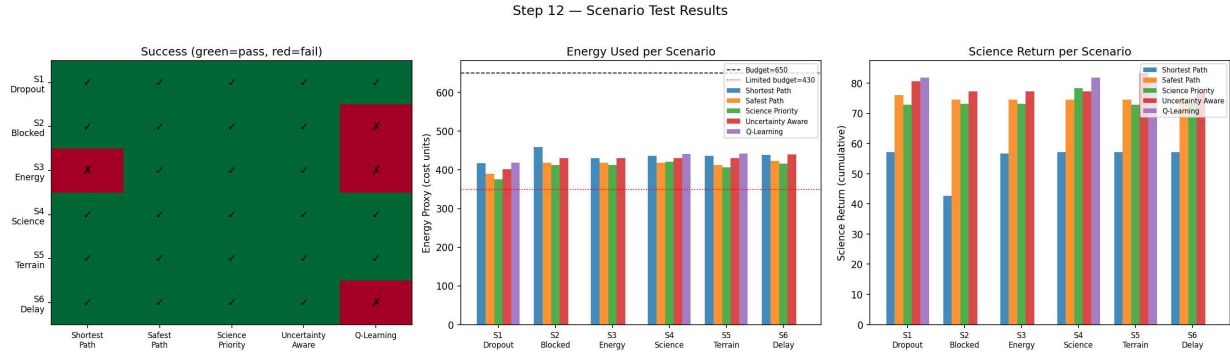


Figure 4: Scenario Test Results. Left: success matrix (green = pass, red = fail). Centre: energy used per scenario per policy. Right: science return per scenario per policy. The red dotted line in the energy panel marks the reduced budget of 430 for Scenario S3.

Table 4: Scenario Test Success Summary

Scenario	Description	Shortest	Safest	Sci. Priority	Unc. Aware	Q-Learning
S1	Sensor Dropout (25%)	Pass	Pass	Pass	Pass	Pass
S2	Blocked Path	Pass	Pass	Pass	Pass	FAIL
S3	Limited Energy (430)	FAIL	Pass	Pass	Pass	FAIL
S4	Wrong Resource Estimate	Pass	Pass	Pass	Pass	Pass
S5	Terrain Uncertainty	Pass	Pass	Pass	Pass	Pass
S6	Communication Delay	Pass	Pass	Pass	Pass	FAIL
	Success Rate	66.7%	83.3%	83.3%	83.3%	50.0%

Key scenario observations:

- S1 (Sensor Dropout): All five policies passed. A* policies replanned using reduced confidence values. Q-Learning followed its frozen policy but the blocked cells happened not to be on its stored path in this scenario.
- S2 (Blocked Path): The four A* policies detected the new obstacle during replanning and found alternative routes. Q-Learning failed because its stored policy map directed it into blocked cells with no recovery mechanism.

- S3 (Limited Energy): Safest (418.9), Science Priority (412.0), and Uncertainty Aware (429.9) all completed within the 430-unit limit. Shortest Path (435.8) and Q-Learning (440.5) both exceeded 430 and failed. The margin for Uncertainty Aware was 0.1 units.
- S4 (Wrong Resource Estimate): All policies passed. Science Priority deviated from its nominal path to visit noise-inflated cells, returning slightly different science values than in the nominal run.
- S5 (Terrain Uncertainty): All five policies were robust to terrain perturbation. Recovery rates were 99 to 101% of nominal performance.
- S6 (Communication Delay): Q-Learning failed under the communication delay scenario because the overhead steps pushed total energy beyond the budget. The three energy-efficient A* policies passed. Shortest Path passed because the delay overhead was within its remaining energy margin.

10. Discussion

10.1 What Worked Well

The A* framework proved robust and flexible. All four baseline policies produced valid paths in the nominal run and handled sensor dropout and terrain uncertainty without failure. The modular cost function design allowed meaningful differentiation between policies: Safest Path and Science Priority both achieved 83.3% success rates despite optimising for different objectives. The Uncertainty Aware policy demonstrated an interesting emergent result — highest science return among all policies — suggesting that in this study area, high-confidence terrain is spatially correlated with high-science-reward terrain.

Q-Learning training converged cleanly at 100% win rate and produces an extremely compact policy representation (39 KB) that is ideal for an onboard-constrained rover. Its $O(1)$ lookup time of 0.76 microseconds is orders of magnitude faster than any A* replanning step.

10.2 What Did Not Work Well

Q-Learning failed two of the six scenarios (S2 and S3) and a third (S6), giving it the lowest overall resilience. The core limitation is that a frozen Q-table cannot adapt to environmental changes. When a path segment is blocked, the learned policy continues directing the rover toward impassable cells and the episode terminates as a failure.

The science return metric is not directly comparable between A* and Q-Learning policies. A* science is the cumulative sum over all path cells. Q-Learning science is measured only at first-visit cells above a 0.5 threshold, deliberately excluding science from training to prevent loitering. This is a methodological inconsistency that should be standardised in future work.

The terrain normalisation denominator inconsistency (slope/25 in code vs slope/15 in documentation) means that all terrain risk values are approximately 40% lower than the documented formula would produce. This may affect the relative weighting between terrain, shadow, and confidence in the movement cost. Standardisation is needed before digital twin integration.

The `study_area_crop_info.txt` metadata file was inherited from an earlier 80 x 80 grid run and does not reflect the current 200 x 200 grid. This file must be regenerated.

10.3 Physical Interpretation

The result that Safest Path uses least energy while taking the longest physical path is physically meaningful. On steep south-polar terrain, the dominant cost is the terrain penalty in the movement cost formula (weight 0.6). A policy that avoids steep slopes accumulates lower per-step energy cost even if it takes more steps. This is consistent with real rover power modelling, where climbing steep slopes at low speed consumes disproportionately more energy per metre travelled.

The Uncertainty Aware result suggests that radar-covered terrain in this study area tends to be flatter and less permanently shadowed, making it more traversable and more likely to contain accessible science targets.

10.4 Integration Readiness

The simulation outputs are spatially registered to a 200 x 200 grid at 25 m per cell resolution. All layer definitions, thresholds, and weight values are documented in this report and in the output files, making the simulation ready for integration with the digital twin pipeline once the terrain normalisation inconsistency is resolved and the study area metadata is regenerated.

11. Integration with Digital Twin

Table 5 lists the simulation outputs produced by this project and their intended downstream use within the broader digital twin framework.

Table 5: Outputs and Integration Use

Output Produced	File	Format	Spatial Unit	Later Use in Digital Twin
Terrain risk map	terrain_risk_map.png + .npz	PNG / numpy array	200x200 @ 25m	Input to traversability layer; feed to path planning module
Shadow risk map	shadow_risk_map.png + .npz	PNG / numpy array	200x200 @ 25m	Input to power/thermal risk layer; feed to mission timing module
Confidence map	confidence_map.png + .npz	PNG / numpy array	200x200 @ 25m	Input to sensor fusion confidence layer (Intern 2)
Science reward map	science_reward_map.png + .npz	PNG / numpy array	200x200 @ 25m	Input to science target prioritisation module (Intern 3)
Baseline policy results	baseline_policy_results.csv	CSV	Per-path metrics	Benchmark comparison for upgraded policies
Q-Learning policy map	q_policy_map.png + .npz	PNG / numpy array	200x200 cells	Onboard action lookup table; deploy as uint8 array
Scenario results	scenario_results.csv	CSV	Per-scenario metrics	Mission risk assessment and policy selection support
Policy compute cost	policy_compute_cost_comparison.csv	CSV	Per-policy metrics	Resource allocation planning for onboard compute budget

Main assumptions: All layers assume the 200 x 200 crop is geographically aligned across datasets. The science reward proxy (radar CPR and S1) is provisional and has not been validated against ground truth composition data. The energy budget (650 units) is a simulation parameter, not a hardware-verified specification.

Main limitations for integration: The terrain normalisation inconsistency must be resolved. The study area metadata file must be regenerated. The Q-Learning policy cannot be used as a standalone onboard system without a fallback replanning mechanism for obstacle encounters.

12. Conclusion

6. A 200 x 200 grid-world simulation was constructed at 25 m per cell resolution from four real lunar south-polar raster datasets (LOLA, LROC, DFSAR, MiniRF), representing a 5 km x 5 km study area with 34,624 passable cells (86.6%) and an energy budget of 650 cost units.
7. Four A* baseline policies and one Q-Learning policy were implemented and evaluated. Safest Path, Science Priority, and Uncertainty Aware all achieved 83.3% success across six mission failure scenarios. Shortest Path achieved 66.7%. Q-Learning achieved 50%, limited by its inability to replan in response to blocked paths or energy overruns.
8. The Safest Path policy is the most resilient A* option: it achieves the lowest energy consumption (418.9 cost units) despite taking the longest path (273 steps), because it avoids high-slope cells that carry the dominant movement cost penalty.
9. Q-Learning produces the most computationally efficient onboard policy (O(1) lookup, 39 KB, 0.76 microseconds per step) but requires a replanning fallback mechanism to be mission-deployable.
10. Three implementation issues were discovered and corrected during the project: disconnected terrain (slope threshold raised from 15 to 20 degrees), science bonus threshold too restrictive (lowered from 99th to 95th percentile), and Uncertainty Aware cost function bug (confidence penalty weight corrected). Two residual inconsistencies remain: terrain normalisation denominator discrepancy and outdated study area metadata file.
11. Recommended next step: implement a hybrid policy that uses the Q-Learning map for nominal execution and switches to a lightweight A* replanner when blocked paths or energy budget violations are detected. This would combine O(1) onboard efficiency with A*-class resilience.

13. References

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Appendix

A. Output File List

Table A1: Output Files Produced by This Project

File	Description	Section
terrain_risk_map.png / .npy	Terrain risk layer at 25m/cell	Step 5
shadow_risk_map.png / .npy	Shadow risk layer at 25m/cell	Step 5
confidence_map.png / .npy	Data confidence layer at 25m/cell	Step 5
science_reward_map.png / .npy	Science reward layer at 25m/cell	Step 5
grid_world.csv	Full grid cell properties	Step 6
grid_cell_properties.csv	Per-cell feature table	Step 6
action_space.txt	Action definitions and direction vectors	Step 7
reward_function.txt	Reward function specification	Step 8
operating_constraints_table.csv	Rover operating constraint table	Step 8
baseline_policy_results.csv	Nominal run metrics for 4 A* policies	Step 9
baseline_policy_maps.png	4-panel path map on terrain risk	Step 9
q_learning_summary.csv	Q-Learning training and evaluation metrics	Step 10
q_policy_map.png / .npy	Learned greedy action map (uint8)	Step 10
policy_compute_cost_comparison.csv	Compute cost and storage for all policies	Step 11
scenario_results.csv	Pass/fail and metrics for all 6 scenarios	Step 12
resilience_metrics.csv	Aggregated resilience metrics per policy	Step 13
scenario_plots.png	Scenario success, energy, and science plots	Step 12
reference_rover_architecture.md	Reference rover assumptions	Step 1
final_report.docx / .pdf	This report	Step 15
final_presentation.pptx	Presentation slides	Step 15

B. Key Simulation Parameters

Table A2: Simulation Parameters

Parameter	Value	Notes
Grid size	200 x 200 cells	40,000 total cells
Cell resolution	25 m/cell	5 km x 5 km total area
Start cell	(0, 199)	Upper-left region
Goal cell	(199, 0)	Lower-right region
Passability threshold	slope <= 20 degrees	Corrected from initial 15 degrees
Impassability threshold (IMPASS_THR)	0.90	Terrain risk >= 0.90 => impassable

Parameter	Value	Notes
Passable cells	34,624 (86.6%)	After slope ≤ 20 degree correction
Energy budget (nominal)	650 cost units	Single-traverse allocation
Energy budget (S3 scenario)	430 cost units	Reduced for limited energy test
Actions	8 (4 cardinal + 4 diagonal)	Diagonal cost multiplier = $\sqrt{2}$
Movement cost weights	$tr=0.6$, $sh=0.3$, $(1-cn)=0.1$	Weights sum to 1.0
Science threshold	95th percentile of passable cells	Corrected from initial 99th percentile
Q-Learning episodes	150,000	
Q-Learning alpha	0.1	Learning rate
Q-Learning gamma	0.99	Discount factor
Q-Learning training time	300 seconds	On project laptop
Q-table size	320,000 entries (1,250 KB)	Full Q-table; offline only
Compact policy map size	39 KB	uint8 argmax array; onboard deployable